Semantic Analysis of Domain Data for Proactive Identification of Cybersecurity Indicators

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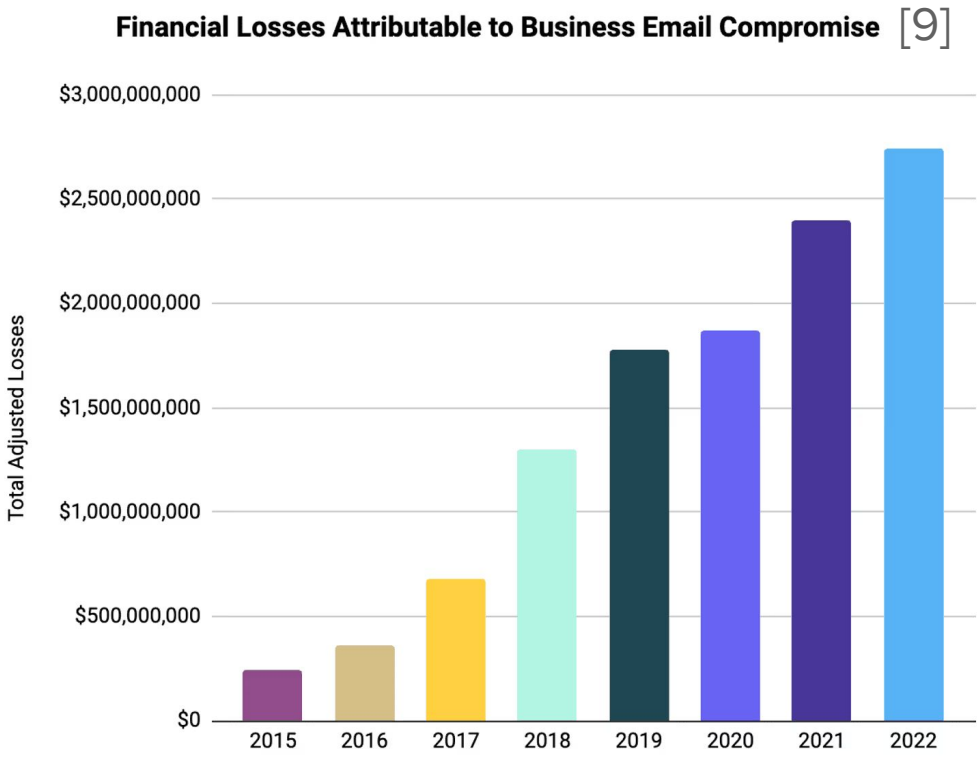
Introduction to Our Topic

- Phishing and Spam wastes resources and harms end users
- Threats rise while detection stagnates
- AI and Data Science abilities have evolved since early Spam filters were made
- Many avenues exist to expand the old guard of solutions



Overall Literature Overview

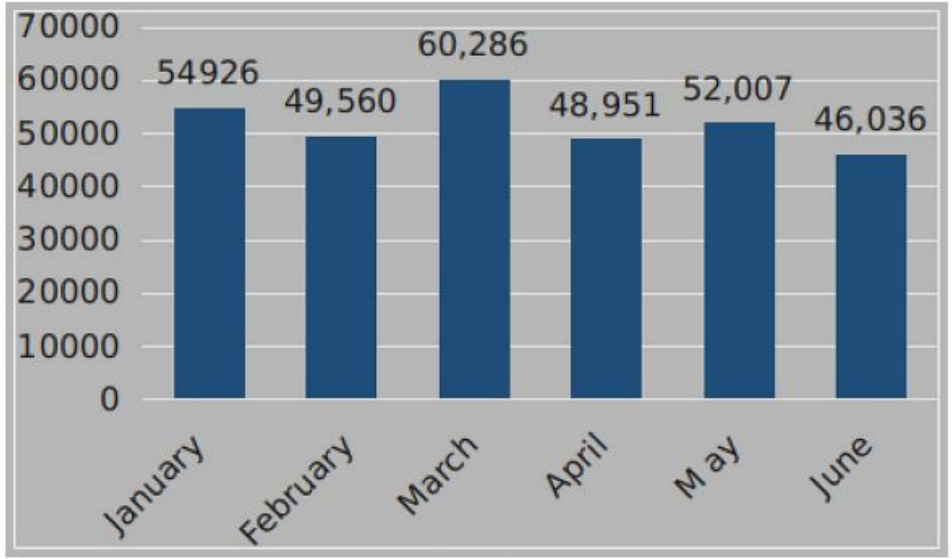
The primary categories of interest for our topic were:

- Reactive Detection of Attacks
- Proactive Data Gathering
- Proactive Prevention of Attacks
- Psychological Studies

Review of Key Papers - Reactive Detection of Attacks

- Major contributors include 2020's *A semantic-based classification approach for an enhanced spam detection* [5] and *Detecting Phishing SMS Based on Multiple Correlation Algorithms* [6]

Fig. 1 Number of unique phishing Web sites detected [6]



Review of Key Papers - Proactive Data Gathering

- 2019's *iMCircle: Automatic Mining of Indicators of Compromise...* [7] and 2023's *Phishing or Not Phishing? A Survey...* [8] both push beyond reactionary remediation

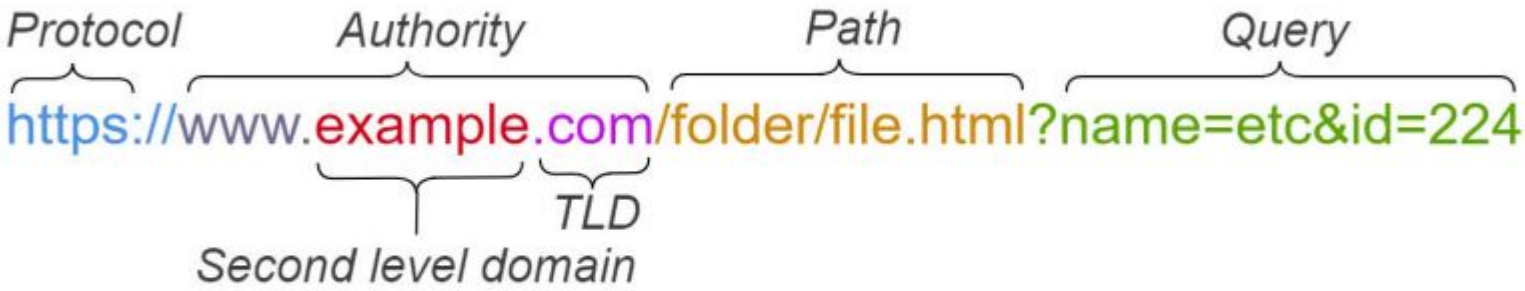


FIGURE 4. Example of a URL with its components. [8]

Review of Key Papers - Proactive Prevention of Attacks

- 2025 brought us *Phish Fighter: Self Updating Machine Learning…* [1] and *APOLLO: A GPT-based tool…* [4] as preventative and semantic tools following the boom of LLMs
- Proactive remediation costs more, but the cost of a single attack can be many orders of magnitude more [9]
- Despite this, the evolution of proactive capabilities have lowered cost while also increasing effectiveness

Review of Key Papers - Psychological Reviews

[3] **TABLE 5** Correlations with actions selected for phishing emails.

- And in a more foundational sense, the 2022-2023 papers of An Effective Cybersecurity Awareness Training Model… [2] and Which phish is captured in the net?… [3] show surveys on how the previously mentioned processes actually affect people

Measure	Follow request	Investigate	Flag	Ignore/delete
d'	−0.45**	−0.16	0.17*	0.33**
c	−0.03	−0.10	0.00	0.09
Hits	−0.42**	−0.12	0.17*	0.29**
FAs	0.45**	0.18*	−0.18	−0.34**
RT	−0.32**	−0.01	0.17*	0.13
Confidence	−0.20*	−0.39**	0.18*	0.32**
PAQ	0.10	−0.18*	0.10	−0.01
Age	−0.33**	−0.14	0.11	0.27**
Big 5-E	0.08	−0.07	−0.08	0.04
Big 5-A	−0.43**	−0.30**	0.18*	0.42**
Big 5-C	−0.37**	−0.30**	0.19*	0.37**
Big 5-N	0.29**	0.21*	−0.13	−0.28**
Big 5-O	−0.05	−0.19*	0.03	0.16*
Impulsivity	0.47**	0.28**	−0.22**	−0.41**
Curiosity-SC	0.27**	0.06	−0.10	−0.18
Curiosity-DS	0.22**	0.03	−0.12	−0.10
Curiosity-ST	0.36**	0.21**	−0.10	−0.35**

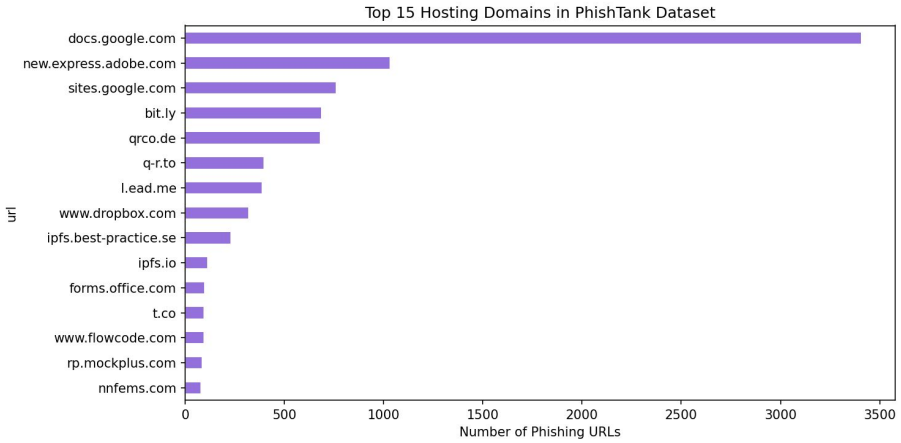
*$p < .05$; **$p < .01$.

Primary Research Gaps Identified

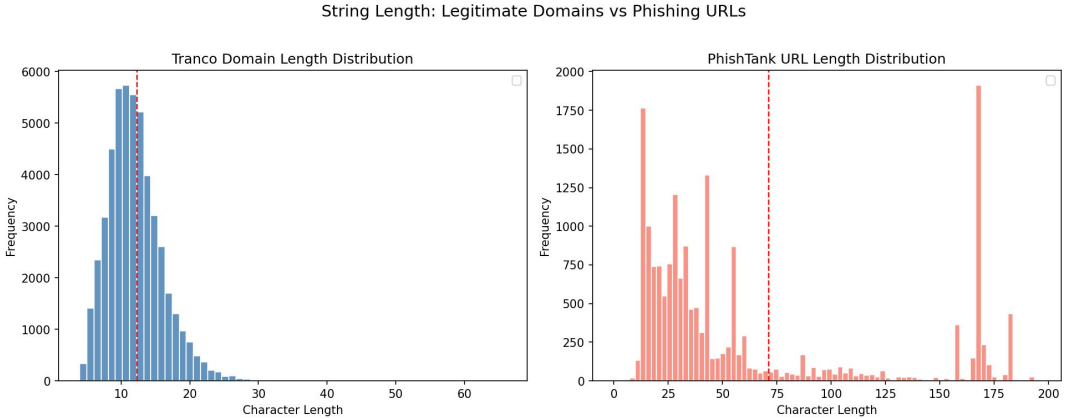
- The Zero-Day Challenge
  - Current systems struggle with novel threats.
  - Detection accuracy degrades rapidly without prior threat data.
- Evolution of ML Capabilities
  - Shift from manual feature engineering to semantic LLM reasoning.
- Empowering the End User
  - Focus on easily interpretable outputs to assist human decision-making

How Our Project Will Build Upon Existing Work

- **Domain Focused Semantic Analysis**
- **Tranco Top 50K** Domains & **Phishtank** Verified Phishing URLs
- Semantic Understanding via **Gemini Text Embeddings** & Comparisons via **Cosine Similarity**
- **Neither dataset is infallible**
- Focus on **Actionable Intel**

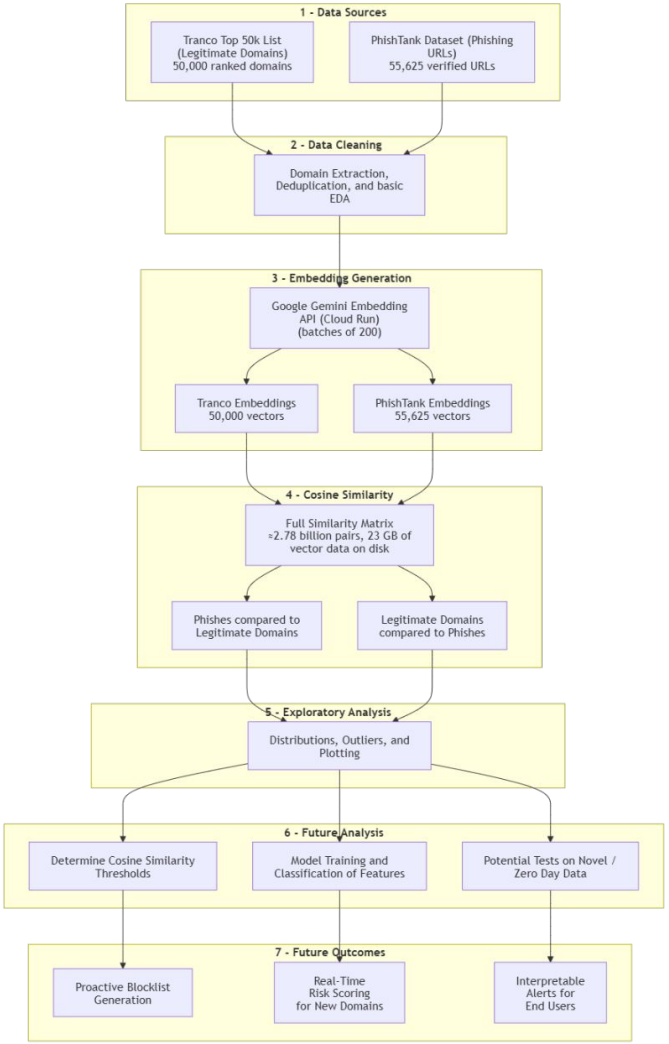



```
                DATA SHAPE
----------------------------------------
  Avg PhishTank URL length      : 71.2
  Avg PhishTank host length     : 20.2
  Avg Tranco domain length      : 12.3
   Median PhishTank URL length   : 38
   Median Tranco domain length   : 12
  Mean hyphens — PhishTank      : 0.93
  Mean hyphens — Tranco         : 0.09
 Mean digit ratio — PhishTank : 0.1023
 Mean digit ratio — Tranco     : 0.0124
  PhishTank unique hosts        : 8,271
 Tranco unique domains         : 50,000
```



Summary & Conclusion

- Semantic Analysis is likely to become instrumental in fighting future attacks
- Embeddings are becoming more and more reasonable to be done locally
- Plenty of unforeseen data features and insights lie in the semantic metadata of the average domain



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